

SingleRAN

Transmission Security Overview Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the support for obtaining CRLs over HTTP. For details, see 4.2 On an Untrusted Network .	Modified parameter: Added the enumerated value HTTP to the CRLTSK.CRLGETMET HOD (LTE eNodeB, 5G gNodeB) parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

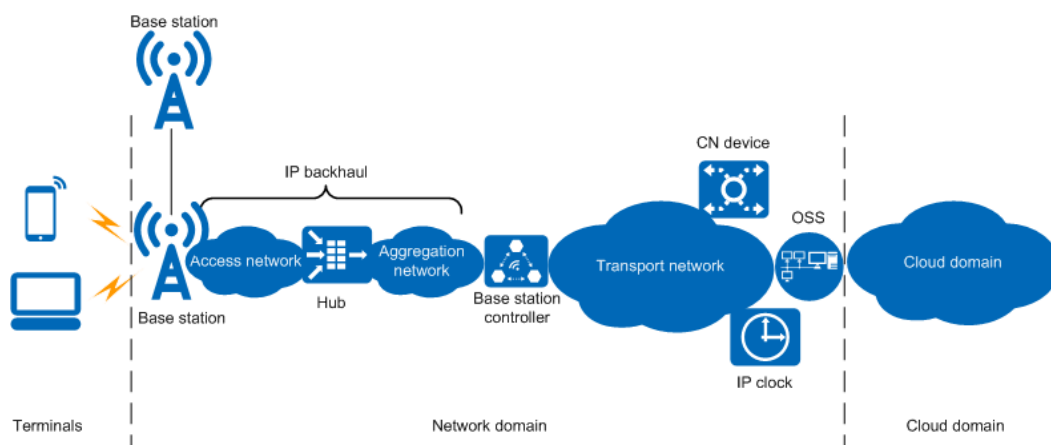
For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

3 Transport Network Overview

3.1 End-to-End Networking for Transmission

Figure 3-1 shows an end-to-end network for transmission. The following describes the transmission security solutions of the IP backhaul network shown in the figure.

Figure 3-1 End-to-end network for transmission



3.2 Evolution

In TDM, ATM, or IP over E1 transmission mode, the transport network is generally dedicated to transmitting radio services. Transmission links are designed to provide secure transmission, without requiring additional security features. However, as the mobile broadband (MBB) network develops, transport networks have evolved into all-IP-based systems, which are completely open and easily accessible. As a result, transport networks transmitting telecommunication services are subject to various security risks.

NOTE

This document only describes transmission security pertaining to Ethernet or IP networks.

Multi-plane security measures are required to protect radio equipment against security threats and malicious attacks, and to provide secure communication over transport networks.

3.3 Security Requirements

As indicated in 3GPP TS 33.210, Network Domain Security for IP based protocols (NDS/IP) is recommended for transmission security.

3.3.1 NDS/IP Dimensions Defined by 3GPP

3GPP defines the following Network Domain Security for IP based protocols (NDS/IP) dimensions:

- Data integrity
Data integrity ensures that data is correct. This protects data from unauthorized modification, removal, or creation and indicates such unauthorized activities.
- Data origin authentication
This ensures that the origin of the data received is as claimed.
- Anti-replay protection
This is a special type of integrity protection that safeguards packets from being intercepted, modified, and then reinserted by a third party (man-in-the-middle).
- (Optional) Data confidentiality
This prevents eavesdropping by allowing only authorized entities to access and parse data.

3.3.2 NDS/IP Mechanisms Defined by 3GPP

From the perspective of NDS/IP, all network nodes are IP nodes. NDS/IP on a 3GPP network uses standard security procedures and mechanisms defined by IETF.

The network is divided into different security domains. These domains are isolated by security gateways (SeGWs). The SeGWs perform routing and implement security policies for inter-domain traffic.

- Each security domain can be configured with multiple SeGWs to avoid a single point of failure (SPOF) or improve the performance by load balancing.
- IPsec authenticates the data source, checks data integrity, and ensures confidentiality.
- A security network is set up as follows:
 - The base station establishes an IPsec tunnel.
 - The IPsec tunnel is set up across the IP backhaul network.
 - The SeGW on the core network terminates the IPsec tunnel.
- The base station uses the public key infrastructure (PKI) or pre-shared key (PSK) to authenticate the identity of the peer end.

4 Transmission Security Solutions

This chapter describes recommended transmission security solutions that meet transmission security standards and operator requirements.

4.1 On a Trusted Network

A trusted network contains physically safe sites. Each site or transport network is managed by a single organization and the operator running a site can strictly control access to it.

On a trusted network, strong authentication is required for accessing a physical network.

The transmission security solutions for a trusted network are as follows:

- Secure Sockets Layer (SSL)
SSL is used to secure all data transmitted through the operation and maintenance (OM) channel between the base station and the OSS or local maintenance terminal (LMT).
- Access control based on 802.1x
The base station is authenticated based on 802.1x before it connects to the network.

Figure 4-1 shows the logical networking for transmission security on a trusted network.

Figure 4-1 Logical networking for transmission security on a trusted network

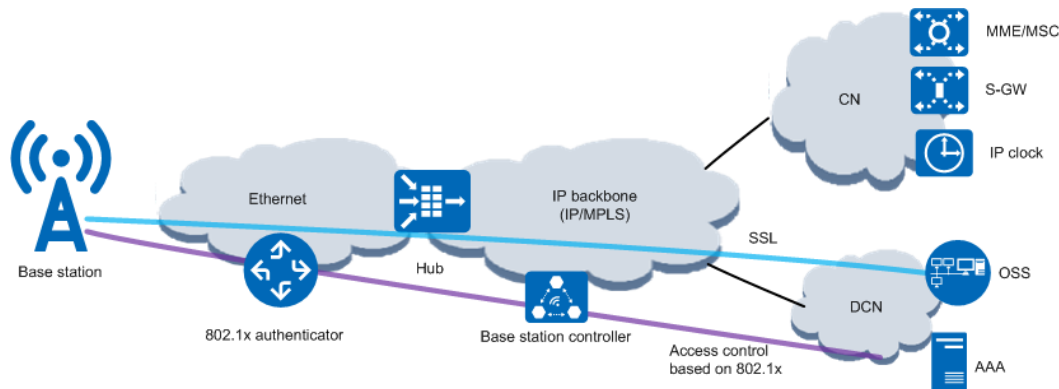


Table 4-1 describes the NEs involved in the transmission security solution.

Table 4-1 NEs involved in the transmission security solution for trusted networks

NE	Description
Base station	Supports SSL and access control based on 802.1x.
MAE	Configures and manages base stations.
Authentication, Authorization and Accounting (AAA) server	Uses digital certificates to implement access control based on 802.1x for base stations.
802.1x authenticator	Implements identity authentication based on 802.1x, which can be a switch on the transport network.

Table 4-2 describes the external interfaces involved in the transmission security solution.

Table 4-2 External interfaces involved in the transmission security solution for trusted networks

External Interface	Description
SSL interface	An interface used to set up an SSL connection between a base station and the MAE
802.1x interface	An interface used by a base station to implement access control through an 802.1x authenticator

4.2 On an Untrusted Network

An untrusted network contains physically unsafe sites. The sites or transport network may be managed by multiple organizations and the operator running a site cannot strictly control access to it.

On an untrusted network, IPsec and other security features are used to protect network access and the data on the user, control, and management planes.

The transmission security solutions for an untrusted network are as follows:

- IPsec
In IPsec networking, a SeGW is deployed to terminate an IPsec tunnel on the core network (CN) side. In addition to the IPsec tunnel solution, IPsec also provides secure base station deployment and IPsec reliability solutions.

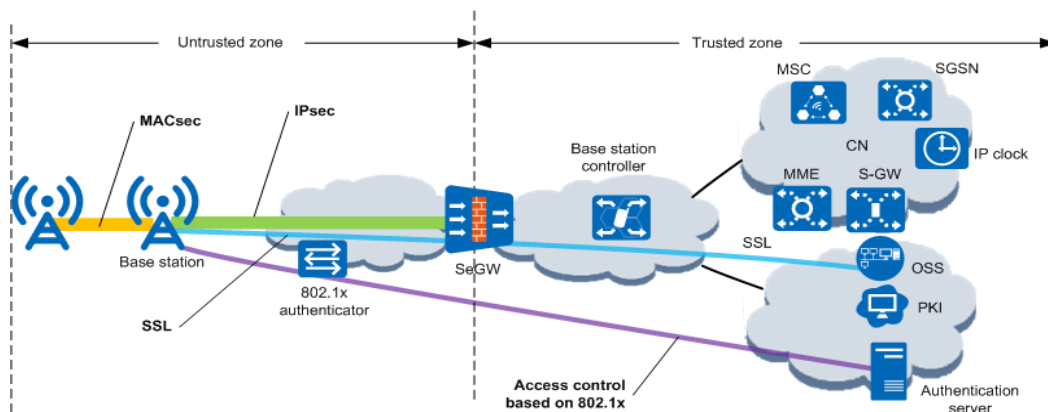
NOTE

Clock packets are carried on one of the user, control, and management planes, indicating that clock packets can be transmitted using the IP address of the user, control, or management plane of the base station.

- **PKI**
The PKI system works with the base station to issue and manage certificates for transmission security authentication during implementation of IPsec, access control based on 802.1x, SSL, and DTLS. The base station supports initial certificates and complies with Certificate Management Protocol v2 (CMPv2).
- **SSL**
SSL is used to secure all OM data transmitted through the OM channel between the base station and the OSS or LMT.
- **Access control based on 802.1x**
The base station is authenticated based on 802.1x before it accesses the network.
- **Media Access Control security (MACsec)**
MACsec is used to protect data transmitted between interconnected UMPT boards.
- **DTLS (NR)**
DTLS is introduced to ensure the security of data transmission based on unreliable protocols, such as UDP. It is recommended that DTLS be used for signaling transmission over the NG-C and Xn-C interfaces to ensure security.

Figure 4-2 shows the logical networking for transmission security on an untrusted network.

Figure 4-2 Logical networking for transmission security on an untrusted network



- **Table 4-3** describes the NEs involved in the transmission security solution.

Table 4-3 NEs involved in the transmission security solution for untrusted networks

NE	Description
Base station	Uses an integrated firewall to prevent attacks and supports VLAN configuration to isolate user, control, and management planes. If two base stations are connected to each other through the Ethernet ports on UMPT boards in two BBUs, the MACsec function protects data transmitted between these Ethernet ports.
MAE	Configures and manages base stations.
AAA server	Uses digital certificates to implement access control based on 802.1x for base stations.
802.1x authenticator	Implements identity authentication based on 802.1x, which can be a switch on the transport network.
SeGW	Terminates an IPsec tunnel and uses an integrated firewall to prevent attacks to the CN.
PKI	Includes the certificate authority (CA), registration authority (RA), and certificate revocation list (CRL) server and manages digital certificates for NEs such as base stations and SeGWs.

- **Table 4-4** describes the external interfaces involved in the transmission security solution.

Table 4-4 External interfaces involved in the transmission security solution for untrusted networks

External Interface	Description
SSL interface	An interface used to set up an SSL connection between a base station and the MAE
802.1x interface	An interface used by a base station to implement access control through an 802.1x authenticator
IPsec interface	An interface used to set up an IPsec tunnel between a base station and SeGW
PKI interface	A CMPv2 interface, by which a base station sends requests to a CA or RA to apply for, revoke, or update a digital certificate An LDAP/FTP/HTTP interface, by which a base station downloads CRLs from a CRL server
UMPT interconnection interface	An interface used to interconnect base stations

4.3 Application Constraints

4.3.1 Scenario 1: RAN Sharing

When RAN Sharing is in use, multiple IPsec tunnels must be established in order to isolate and protect the data of each operator.

As of SRAN10.1, different operators can use different certificates.

4.3.2 Scenario 2: Transmission on Public Networks

For transmission on public networks, IPsec tunnels must support network address translation (NAT).

4.3.3 Scenario 3: Cascaded Base Stations

When base stations are cascaded, each base station must be safeguarded using IPsec. It is recommended that each base station possess an individual IPsec tunnel and that the hub base station only perform forwarding.

4.3.4 Scenario 4: Interconnected Base Stations

MACsec is used to protect data transmitted between interconnected UMPT boards in two base stations.

4.3.5 Scenario 5: Networking That Does Not Support the Transmission Security Solution

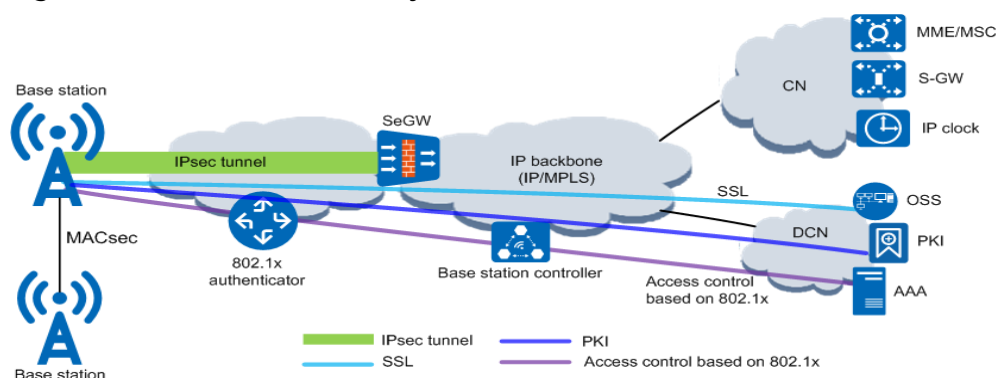
If an FDD eNodeB is configured with two UMPT boards, IPsec tunnels using certificate-based authentication can be established only between one UMPT board and an SeGW.

5 Transmission Security Features

5.1 Introduction

Transmission security features include IPsec, access control based on 802.1x, SSL, PKI-CMPv2, MACsec, and DTLS, as shown in [Figure 5-1](#).

Figure 5-1 Transmission security features



5.2 IPsec

Defined by the IETF, IPsec is a security framework that provides secure end-to-end data transmission on untrusted networks, such as the Internet. On IP networks, IPsec provides transparent, interoperable, and cryptography-based security services to ensure the confidentiality, integrity, and authenticity of transmissions and to provide anti-replay protection.

IPsec operates at the IP layer of the TCP/IP protocol stack and provides transparent security services for upper-layer applications.

For details, see *IPsec for SingleRAN*.

5.3 Access Control Based on 802.1x

802.1x is a standard for port-based network access control. Access control based on 802.1x involves a client (base station), authentication access equipment (LAN switch), and authentication server (AAA server).

During initial connection of a base station to the network, only 802.1x authentication packets can be transmitted because a base station authentication procedure is not completed. After the authentication server authenticates the base station and authorizes the port, normal data transmission is allowed on the authorized port. This ensures that only authorized users are allowed to access the network, protecting the security of the network.

For details, see *Access Control based on 802.1x* for SingleRAN.

5.4 SSL

SSL is a security protocol developed by Netscape. It aims to provide authentication, confidentiality, and integrity protection for two communication applications. The latest standard version is Transport Layer Security version 1.3 (TLS1.3).

SSL enables an end-to-end secure connection to be established between two pieces of equipment. The details are as follows:

- SSL operates between the TCP and application layers. It is established over reliable transport layer protocols but operates independently from application layer protocols.
- Before any communication using application layer protocols occurs, encryption algorithm negotiation, key negotiation, and server authentication must be completed.
- Application layer protocols such as HTTP, FTP, and Telnet can be transparently established over SSL. All data transmitted using the application layer protocols is encrypted to ensure confidentiality.

SSL is also established between the base station or base station controller and the MAE to protect OM data and provide secure remote maintenance.

For details, see *SSL* for SingleRAN.

5.5 PKI

PKI uses an asymmetric cryptographic algorithm to provide information security. PKI manages keys and digital certificates that are essential for services such as encryption and digital signature on all networks. The functionalities and interfaces related to PKI comply with X.509 and 3GPP TS 33.310.

A PKI system consists of a CA, a certificate & CRL database, end entities, and an optional RA.

PKI defines a certificate management system. As a certificate management protocol, CMPv2 is used to exchange certificate management information

between NEs in a PKI system. CMPv2 provides certificate-related, key-related, cross-certification, notification, and other functions. Certificate-related functions include certificate registration, request, and revocation. Key-related functions include key update and recovery. Notification functions include CA key update notification and certificate issuance and revocation notification.

CMPv2-based certificate management is supported by the PKI system. According to the CMPv2 protocol, information such as certificate request, issuance, and update is exchanged between base stations and the PKI system.

For details, see *PKI* for SingleRAN.

5.6 MACsec

MACsec ensures secure communication over an IEEE 802 LAN. MACsec specifically provides user data encryption, data frame integrity check, and data origin authentication at the MAC layer.

MACsec is used by default for base stations to protect data transmitted between interconnected UMPT boards.

For details, see *MACsec* for SingleRAN.

5.7 DTLS (NR)

Public IP networks cannot ensure transmission security like traditional time division multiplexing (TDM) networks. Therefore, when operators use public IP networks to carry wireless services, it is recommended that transmission security protocols be used to protect signaling message transmission. The original Transport Layer Security (TLS) protocol is widely used between the application layer and transport layer to ensure network communication security. However, this protocol requires that the data be carried on a reliable transmission channel, indicating that data is transmitted mainly based on the Transmission Control Protocol (TCP). TLS cannot ensure the security of data transmission based on User Datagram Protocol (UDP), which may experience packet loss and reordering.

With wide applications of UDP, the Datagram Transport Layer Security (DTLS) protocol is introduced. DTLS is applicable to either reliable protocols such as Stream Control Transmission Protocol (SCTP) or unreliable protocols such as UDP. UDP packet forgery is easy. Therefore, DTLS running over UDP is vulnerable to denial of service (DoS) attacks. DTLS is recommended for SCTP, which defends against forged packets. For details, see section 4.1.2.7 "Handling Invalid Records" in IETF RFC 6347.

The DTLS versions include DTLS1.0 and DTLS1.2. According to "Appendix E.1 General" in 3GPP TS 33.310 V15.0.0, DTLS1.0 is not recommended. Therefore, Huawei base stations support only DTLS1.2. The DTLS function of Huawei base stations supports SCTP, which is applicable to the NG-C and Xn-C interfaces.

For details, see *DTLS* in 5G RAN Feature Documentation.

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

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11. *IPsec* for SingleRAN
12. *Access Control based on 802.1x* for SingleRAN
13. *SSL* for SingleRAN
14. *PKI* for SingleRAN
15. *MACsec* for SingleRAN
16. 5G *RANDTLS*